Payload: REASON (Radar for Europa Assessment and Sounding: Ocean to Near-surface)

REASON is a dual-frequency ice-penetrating radar sounder currently flying aboard NASA's Europa Clipper spacecraft. The instrument was developed by researchers at the University of Texas Institute for Geophysics and fabricated by JPL, and it launched on October 14, 2024 atop a SpaceX Falcon Heavy rocket bound for Europa [3]. REASON operates at two frequency bands simultaneously, allowing it to distinguish between near-surface structure and deep basal reflections, a capability that no previous Europa-targeted instrument has demonstrated [1]. The instrument is flight-proven hardware with full mission-level heritage, placing it at TRL 9.

Europa Clipper carries REASON as part of a flyby science campaign: the spacecraft will loop past Europa roughly 50 times, collecting wide-swath regional maps of the ice shell from altitude [2]. However, my mission seeks to address things not covered by the CLIPPER mission. What Clipper is not doing is surface-level, high-resolution traverse profiling targeted at a specific candidate cryobot insertion site. There is no rover, no in-situ localization, and no onboard site-selection logic. The orbital geometry also limits any single flyby to a brief snapshot rather than a continuous along-track profile.

This mission is esentially a continuation of CLIPPER. A surface rover carrying an adapted REASON-heritage sounder drives a continuous traverse, building a spatially registered subsurface map at 100 m resolution or better, and autonomously flags the thinnest viable ice for a future cryobot deployment.

I. **What parameters are measured?**

REASON measures four primary parameters to map Europa's subsurface environment. Two-Way Travel Time (ns) is the elapsed time between the transmitted pulse and the received basal reflection, and serves as the raw measurement. Signal Amplitude (dB) captures the strength of the return from the ice-ocean interface; echoes that fall into the noise floor are flagged to prevent false positives in site selection. Bulk Ice Dielectric Permittivity (dimensionless) is estimated from the sounding's travel characteristics ($\varepsilon_r \approx 3.15$ for pure water ice), with deviations flagging brine pockets or structural impurities. Finally, Derived Ice Shell Thickness (m) is computed autonomously onboard using the relationship $d = c \cdot t \,/\, (2\sqrt{\varepsilon_r})$, where c is the speed of light.

II. **What would a single data entry or record look like?**

REASON produces time-stamped radar trace logs stored as a multi-sheet spreadsheet aligned by trace identifier. Each entry records subsurface sounding parameters at a given rover position, enabling reconstruction of the ice-shell profile along the traverse. The dataset is structured as a spatial series, with rows representing individual radar traces and columns representing measured and derived parameters. Each record includes a trace ID, sol number, rover position in local ENU coordinates, two-way travel time, return amplitude, estimated permittivity, derived ice thickness, accumulated radiation dose, and a quality flag.

Example of a single data record:

On Sol 1 at 94.4 s elapsed mission time (trace TR-0001), REASON recorded a two-way travel time of 21,647.3 ns, a return amplitude of −41.62 dB, a bulk ice permittivity of 3.06, and a derived ice shell thickness of 1,855.5 m at rover position (93.8 m east, −7.1 m north), with an accumulated total ionizing dose of 0.542 krad and the record flagged as NOMINAL.

III. **Who is the intended user of this data?**

There are multiple users of the REASON data, but I will focus on the two most crucial. The onboard autonomous planning algorithm serves as the real-time consumer: it continuously processes daily trace summaries to locate viable cryobot deployment sites (defined as ice thickness <5 km, surface slope $<20°$, and roughness index <0.7) and issues a deployment trigger without human intervention when conditions are satisfied. Earth-bound planetary scientists form the long-term consumer base. The full downlinked dataset provides a high-resolution, spatially registered along-track profile of Europa's ice shell that can be used to model global habitability, constrain ocean-access scenarios, and plan future deep-subsurface entry missions.

References:

[1] NASA Jet Propulsion Laboratory, "Europa Clipper," JPL, Pasadena, CA. [Online]. Available: https://www.jpl.nasa.gov/missions/europa-clipper/. [Accessed: Jun. 16, 2026].

[2] Wikipedia contributors, "Europa Clipper," *Wikipedia, The Free Encyclopedia*. [Online]. Available: https://en.wikipedia.org/wiki/Europa_Clipper. [Accessed: Jun. 16, 2026].

[3] University of Texas at Austin Jackson School of Geosciences, "Scientific Instrument Developed by UTIG Scientists Headed to Europa," UT Austin, Austin, TX, Apr. 2025. [Online]. Available: https://www.jsg.utexas.edu/news/2025/04/on-oct-14-2024-nasas-europa-clipper-success [Accessed: Jun. 16, 2026].